Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/18/2005
PI Name:	Lang, Thomas F. Ph.D.		
Project Title:	Sub-regional Assessment of Bone Loss in the Axial Skeleton in Long-Term Spaceflight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94143-0649	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	10/01/1998	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	Expeditions 2-8 Data collection complete. Awaiting final report at the end of the cooperative agreement.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Leblanc, Adrian (Baylor College of Medicine) Genant, Harry (University of California) Lu, Ying (San Francisco)		
Grant/Contract No.:	NNJ04HC75A		
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Performance Goal Text:			

Task Description:	Bone loss is an established risk factor for long-duration space flight. Experience with the Mir missions and long-term bedrest studies has demonstrated the regional extent of this bone loss, its large variability among individuals, and the fact that it persists despite countermeasures which preserve skeletal loading. Despite this important initial progress, several key gaps in our knowledge remain. There is little information about (1) the distribution of this loss between the trabecular and cortical envelopes of the central skeleton, (2) the cortical/trabecular distribution of the recovery of bone mass post-flight, and (3) the extent to which these changes compare to the spread of bone mineral density (BMD) measures in healthy men and women. The goal of this scientific proposal is to address these three key issues. We will achieve these goals by carrying out a longitudinal study of 10-15 individuals making 4-6 month flights on the International Space Station (ISS). We are imaging these subjects pre-flight, immediately postflight, and 1-year postflight using Quantitative Computed Tomography (QCT) of the spine and hip, dual x-ray absorptiometry (DXA) of the spine, hip and calcaneus, and calcaneal ultrasound. We are assessing sub-regional changes in the spine and the femur by analyzing the QCT images using newly-developed algorithms which compute BMD in trabecular and cortical sub-regions in the spine and hip. We will compute bone loss and recovery in these sub-regions and compare the measurements to the corresponding DXA measures which integrate larger volumes of trabecular and cortical bone. This comparison will establish the extent to which DXA measurements underestimate significant local changes which may impact fracture risk. To compare these changes to the variation of bone density in healthy normal populations, we have acquired a reference database consisting of proximal femoral and spinal QCT scans of healthy Caucasian-American males and females between the ages of 35-45, and are in the process of acquiring a comparable reference data set for African-Americans. Finally, in order to establish the utility of calcaneal ultrasound as a surrogate for calcaneal x-ray absorptiometry, we will compare calcaneal ultrasound and DXA measurements. To characterize the compartmental bone changes in biomechanically relevant anatomic sub-regions of the spine and proximal femur associated with long-duration exposure to microgravity and with re-adaptation to earth gravity. To compare these changes with those quantified by DXA images. To characterize the changes in quantitative ultrasound parameters (broadband ultrasound attenuation and speed of sound) of the heel and compare these to changes measured by DXA of the heel. By quantifying the extent of bone loss during the mission and the extent to which it is recovered after the mission, this protocol yields information on the extent to which individuals who have undergone long-duration spaceflight may be subject to long-term increased skeletal frailty. By defining the time course of skeletal recovery, it will yield information on the frequency with which crewmembers may participate in long-duration missions. By examining differences between integral bone measures such as DXA and sub-regional measurements it has impact on the meaning of the DXA information in terms of tracking bone loss.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary goal of our study is to determine the extent to which the skeletal changes occurring in long-duration human spaceflight are concentrated in small sub-volumes of bone or are more generally distributed throughout the bone. If bone loss is concentrated locally, then measures of bone density (such as DXA, the standard clinical measurement method) which integrate across large volumes of bone may not adequately assess the change in bone strength which occurs in long-duration spaceflight and in the recovery period after the flight. The primary benefits of our ground study derive from the acquisition of a normative database of spinal and femoral QCT scans of healthy Caucasian- and African American males and females between the ages of 35-45. First, by providing reference bone mineral density values against which patient measurements may be compared, this experiment will help facilitate the application of QCT of the hip to diagnosis of osteoporosis in the elderly population. With the increasing availability of new CT scanners in outpatient clinics and small hospitals, management of osteoporosis in the growing elderly population may become an important application of these devices. Additionally, the acquisition of reference data for Caucasian-American males and for African-Americans of both genders, will benefit groups which have not been the main focus of osteoporosis research efforts, which have traditionally centered on Caucasian females. A second benefit is that the QCT reference data comparing African- and Caucasian-Americans will provide important data on gender and racial differences in bone density and geometry in the hip and spine. Almost all of the studies in this area have utilized DXA, which measures an areal density depending on both on the mineral content and the size of the bone. Thus, if two racial groups have different bone sizes, DXA may provide incorrect estimates of the racial difference in bone density. By utilizing QCT, on the other hand, we will be able to measure the volumetric bone mineral density in the hip and spine as well as the three-dimensional geometric properties of these bones.
Task Progress:	As of the end of 2004, we have already completed all of the pre-postflight measurements. 19 crewmembers from Expeditions 2-8 have had pre- and postflight measurements. In summer 2004, we published the pre-postflight results in the Journal of Bone and Mineral Research : Lang T, LeBlanc A, Evans H, Lu Y, Genant H, Yu A, 2004 Cortical and trabecular bone mineral loss from the spine and hip in long-duration spaceflight, J Bone Miner Res. Jun;19(6):1006-12. This publication was based on results from 14 crew on Expeditions 2-6, but the subsequent addition of 4 crewmembers from Expeditions 7 and 8 did not significantly change the results. We are also nearing completion of the Postflight+1year measurements. At this point we only await the 1 year measurements from Expedition 8 for the experiment to be completed. If both crewmembers are studied, this will bring the number of crew with complete measurements (pre-post and 1year) to 17. We expect to submit our findings for publication shortly after completion of the study in April 2005.
Bibliography Type:	Description: (Last Updated: 03/20/2017)
Articles in Peer-reviewed Journals	Lang T, LeBlanc A, Evans H, Lu Y, Genant H, Yu A. "Cortical and trabecular bone mineral loss from the spine and hip in long-duration spaceflight." J Bone Miner Res. 2004 Jun;19(6):1006-12. Epub 2004 Mar 8. PMID: 15125798 , Jun-2004